

17. (2.25 points)

Given the following information for a retrospectively rated book of business:

Policy Effective Year	Ultimate Loss	Losses Reported at Prior Retro Adjustment	Premiums Booked from Prior Adjustment	Premiums Booked as of December 31, 2017
2014	\$226,500	\$202,000	\$335,000	\$337,000
2015	\$225,000	\$180,500	\$330,000	\$335,000
2016	\$200,000	0	0	\$210,000
2017	\$80,000	0	0	\$215,000

Retro Adjustment Period	Selected PDL Ratio	Selected CPDL Ratio	Percentage of Ultimate Loss Reported
First	1.75	???	78.5%
Second	0.70	0.59	10.0%
Third	0.55	0.49	7.0%
Fourth	0.45	0.40	4.0%
Subsequent	0.00	0.00	0.5%

- Retro premium is booked 24 months after policy effective dates.

a. (1.25 points)

Calculate the premium asset as of December 31, 2017.

b. (1 point)

Describe two reasons why the selected PDL ratios decrease as the retro adjustment period increases.

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EXAM 7 SPRING 2018 SAMPLE ANSWERS AND EXAMINER'S REPORT

SPRING 2018 EXAM 7, QUESTION 17

TOTAL POINT VALUE: 2.25

LEARNING OBJECTIVE(S): A14

SAMPLE ANSWERS

Part a: 1.25 points

Sample 1

$$CPDLD1 = \frac{1.75(0.785) + 0.7(0.1) + 0.55(0.07) + 0.45(0.04) + 0(0.005)}{(0.785 + 0.1 + 0.07 + 0.04 + 0.005)} = 1.5$$

Expected Future Loss Emergence:

14: $226,500 - 202,000 = 24,500$

15: $225,000 - 180,500 = 44,500$

16-17: $(200,000 + 80,000) - (0+0) = 280,000$

Expected Future Premium (= Expected Future Loss Emergence × CPDLD)

14: $24,500 \times 0.49 = 12,005$

15: $44,500 \times 0.59 = 26,255$

16-17: $280,000 \times 1.5 = 420,000$

Estimated Total Premium (= Expected Future Premium + Premiums Booked from Prior Adjustment)

14: $12,005 + 335,000 = 347,005$

15: $26,255 + 330,000 = 356,225$

16-17: $420,000 - 0 = 420,000$

Sum of Estimated Total Premium for All Years: 458,260

Premium Asset by Year (= Estimated Total Premium - Premiums Booked as of December 2017)

14: $347,005 - 337,000 = 10,005$

15: $356,225 - 335,000 = 21,225$

16-17: $420,000 - 425,000 = -5,000$

Total Premium Asset = $10,005 + 21,225 - 5,000 = 26,230$

Sample 2

$$CPDLD1 = \frac{1.75 \times (0.785) + 0.7 \times (0.1) + 0.55 \times (0.07) + 0.45 \times (0.04) + 0 \times (0.005)}{(0.785 + 0.1 + 0.07 + 0.04 + 0.005)} = 1.50025$$

Expected Future Premium

14: $226,500 - 202,000 = 24,500 \times 0.49 = 12,005$

15: $225,000 - 180,500 = 44,500 \times 0.59 = 26,255$

16-17: $280,000 - 0 = 280,000 \times 1.50025 = 420,070$

Total = $12,005 + 26,255 + 420,070 = 458,330$

Premium at Prior Retro = $330,000 + 335,000 = 665,000$

Premium Booked = $337,000 + 335,000 + 210,000 + 215,000 = 1,097,000$

Total = $458,330 + 665,000 - 1,097,000 = 26,330$

EXAM 7 SPRING 2018 SAMPLE ANSWERS AND EXAMINER'S REPORT

Part b: 1 point

Sample responses:

- More losses are piercing the individual claim caps/per-accident limit or per-occurrence limits as development continues, which leads to more losses than premium to the insurer, decreasing the PDLD.
- The first period includes basic premium that contemplates certain expenses to be paid once. The remaining PDLD are calculated on an incremental loss emergence basis – so they will decrease as time increases.
- More losses have already been capped in the numerator of the loss capping ratio, but the denominator is uncapped and keeps increasing with a static/ low increasing numerator and a steadily increasing denominator, the PDLD ratios decrease.
- As the period increases, the retro premium is more likely to be impacted by the maximum, so while losses would still be developing, there would be no more premium development since it is at the maximum premium.
- Over time the retro rating parameters of the book of business shifted (e.g. per-accident limit decreases) and PDLD ratios decline.

EXAMINER'S REPORT

Candidates were expected to understand the formula for deriving a retrospective premium asset. Candidates were also expected to demonstrate an understanding of how and why retrospective premium develops relative to losses over time.

Part a

Candidates were expected to calculate the correct premium asset.

Common mistakes included:

- Applying the wrong CPDLD ratios to the policy year reserves.
- Applying the CPDLD ratios to ultimate losses instead of loss reserves.
- Not including the calculation of 2017, and giving a reason for not including as “Policy Year is too recent” or “the amount was less than \$0”.

Part b

Candidates were expected to describe why the PDLD ratios decrease over time.

Common mistakes included:

- Providing incomplete responses (e.g. listing versus describing, or only providing a single response).
- Stating the PDLD ratio decreases because there is less loss development over time.